Air quality, one hour ahead.

Can pollution history and weather improve on simply repeating the latest reading?

43,824
hourly records

4
models compared

+1 hour
forecast horizon

THE PHASE 1 WORKFLOW

01 Literature + problem framing

02 Data audit + training-only EDA

03 Features + four forecasting models

04 Selection + rolling backtests

05 Diagnostics + Aircast demonstration

UCI Beijing · 2010–2014 · PM2.5 concentration in μg/m³

Measured outcome: modest pooled gains, a near tie with Ridge, and unresolved peak errors.

Preserve the clock, the gaps, and the peaks

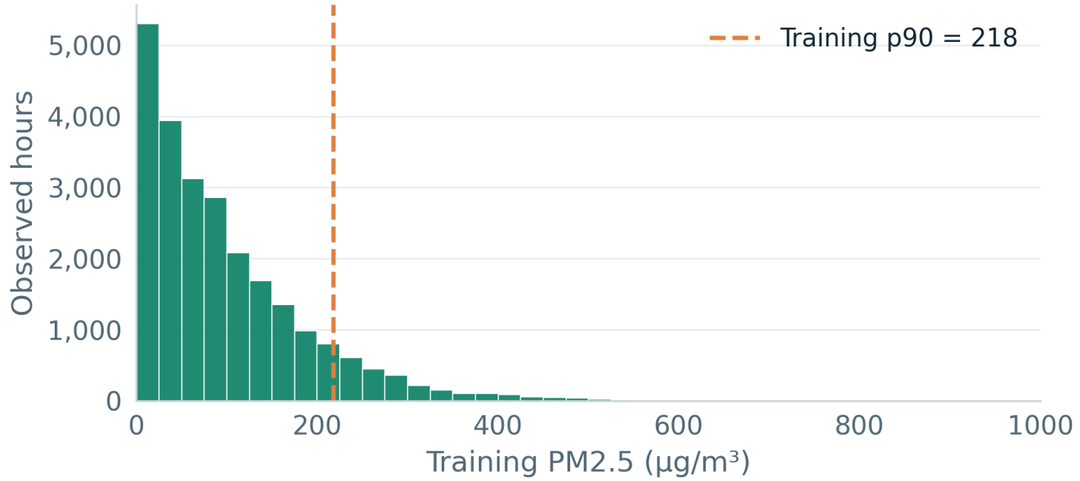
UCI Beijing PM2.5 · January 2010–December 2014 · pollution and airport weather use different sites.

43,824
Regular hourly timestamps
0 duplicates; 0 absent timestamp rows

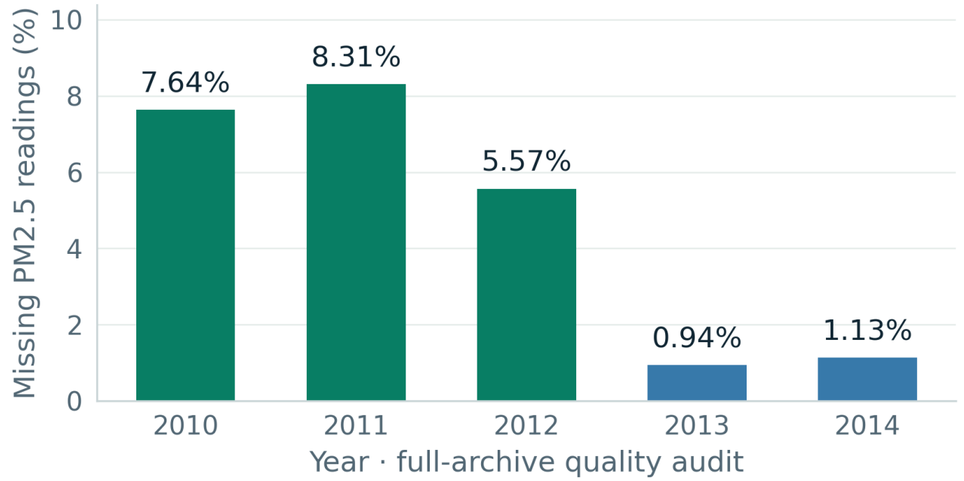
2,067
Missing concentrations
4.72% of the archive; weather complete

155 h
Longest missing run
214 concentration-gap runs in total

Training concentration distribution



Missingness varies by year



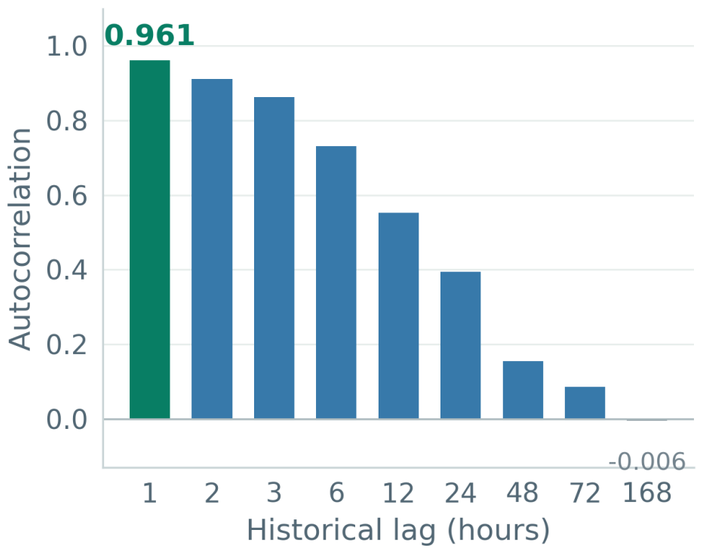
Cleaning: validate schema and dates → preserve hourly slots → retain valid peaks → exclude row ID

SOURCE: UCI 381; supplied CSV audit; concentration distribution uses 2010–2012 only.

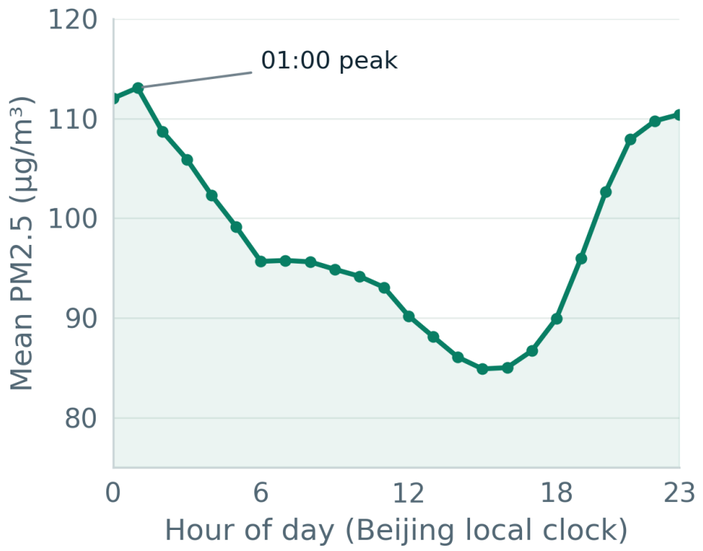
Recent history is the strongest starting point

All three graphs use 2010–2012 only; associations guide candidate features, not causal claims.

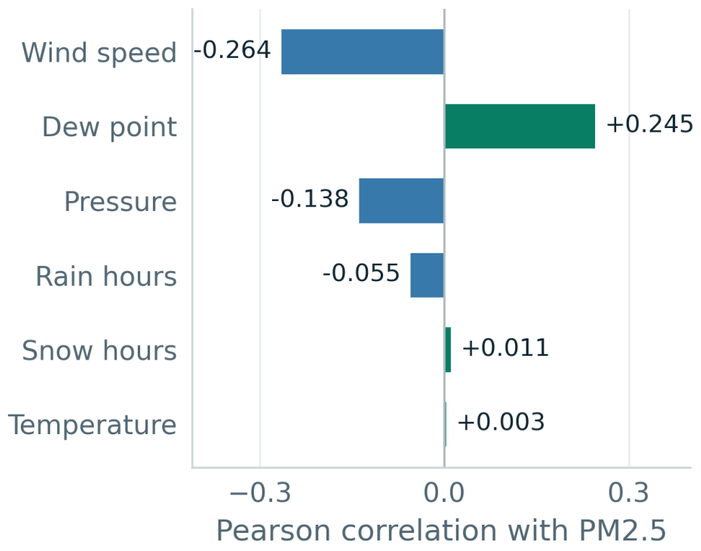
Temporal dependence



Average daily pattern



Weather associations



Take persistence seriously

1-hour ACF 0.961; 24-hour 0.394.
A weekly dependency is not established.

Encode the clock cyclically

The pooled mean peaks at 01:00.
Averages do not predict each event.

Validate the extra information

Wind r = −0.264; dew point +0.245.
Correlation alone does not imply cause.

SOURCE: Training autocorrelation, hourly profiles and Pearson correlations; saved EDA outputs.

Build one forecast row without future information

Predict y(t + 1) from measured history through t · 70 raw columns → 73 with the four observed wind categories.

ORIGIN t = 10:00
Observed PM2.5 + weather + trailing history

→

TARGET t + 1 = 11:00
Only the future calendar is already known

Four feature groups

20 **Pollution lags + missingness**
Offsets 0, 1, 2, 3, 6, 12, 24, 48, 72, 168 h

18 **Trailing statistics**
3, 6, 12, 24, 72, 168 h: mean, std, observed fraction

7 **Known target calendar**
Hour / weekday / month sine–cosine + weekend

25 **Weather + wind direction**
6 weather fields at 0 / 1 / 24 h; flags + wind encoding

Four models, one target

Persistence $\hat{y}(t + 1) = y(t)$
Strong latest-reading reference

Daily seasonal naive $\hat{y}(t + 1) = y(t - 23)$
Same target hour on the previous day

Ridge regression Squared error + L2 penalty (α = 10)
Fold-fitted median imputation and scaling

XGBoost Additive trees + complexity penalty
Native numeric missing splits; outputs clipped ≥ 0

Safeguards: 168-hour warm-up · windows end at t · fold-fitted preprocessing · evaluation targets stay observed

Tune on 2013, then freeze the settings

Quarterly expanding refits · observed histories update hourly within each evaluation block.

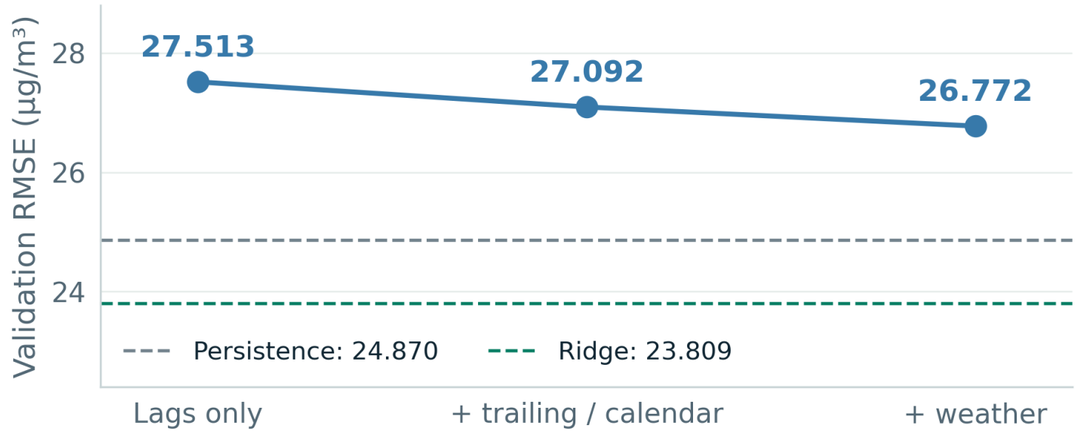
2010–2012
Initial training + EDA

2013
Q1 Q2 Q3 Q4 · validation

2014
Q1 Q2 Q3 Q4 · holdout

Does extra information help XGBoost?

Selected parameters fixed; validation dates only



Features improve XGBoost; simpler validation models remain stronger.

Selection and label-availability rules

Three-candidate grid → lowest pooled validation RMSE

Candidate RMSE: 26.772 / 26.801 / 26.953

Selected: 300 trees · depth 3 · learning rate 0.05

Child weight 5 · row 0.8 / column 0.9 · L2 = 5

For target block B: latest training label ≤ B − 1 h

No holdout tuning or early stopping; fit preprocessing per fold.
Same observed targets and naive inputs for all models.

SOURCE: configs/phase1.toml, selection.json and validation_ablation.csv; rolling-origin forecasting principles.

The gain is useful, but modest

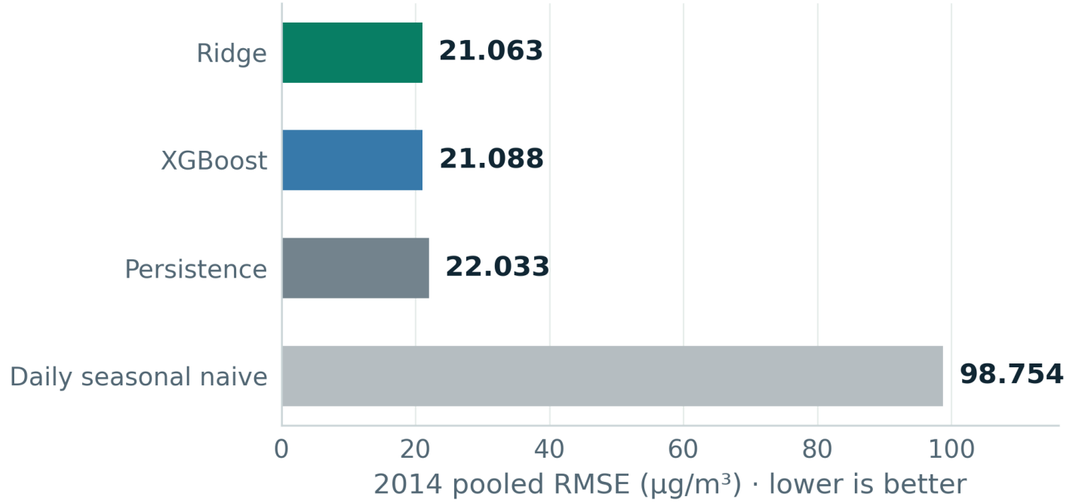
Same 8,531 observed 2014 target hours for all four models · MAE / RMSE in µg/m³; R² is dimensionless.

4.29%
XGBoost RMSE improvement
Relative to persistence, on common hours

4.41%
Ridge RMSE improvement
The linear comparator is slightly better

97.39%
Holdout target coverage
8,531 / 8,760 expected hours; 229 excluded

Annual RMSE comparison



All required metrics

Model	MAE	RMSE	R²
Persistence	11.926	22.033	0.9442
Seasonal naive	67.296	98.754	-0.1205
Ridge	**11.825**	**21.063**	**0.9490**
XGBoost	11.833	21.088	0.9489

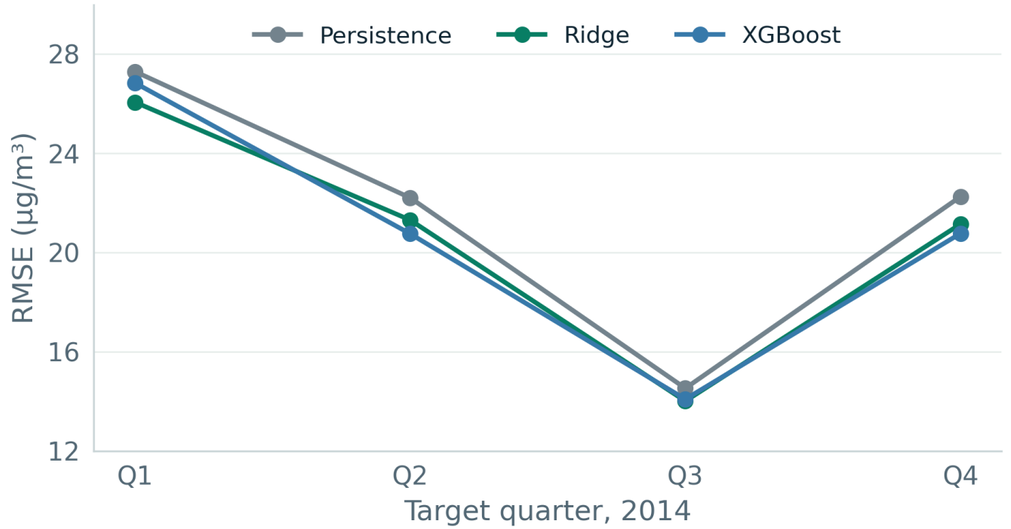
Ridge has lower RMSE by 0.025 µg/m³. The significance of this difference has not been tested.

SOURCE: Real-data metrics.csv; pooled errors recomputed from saved out-of-fold predictions.

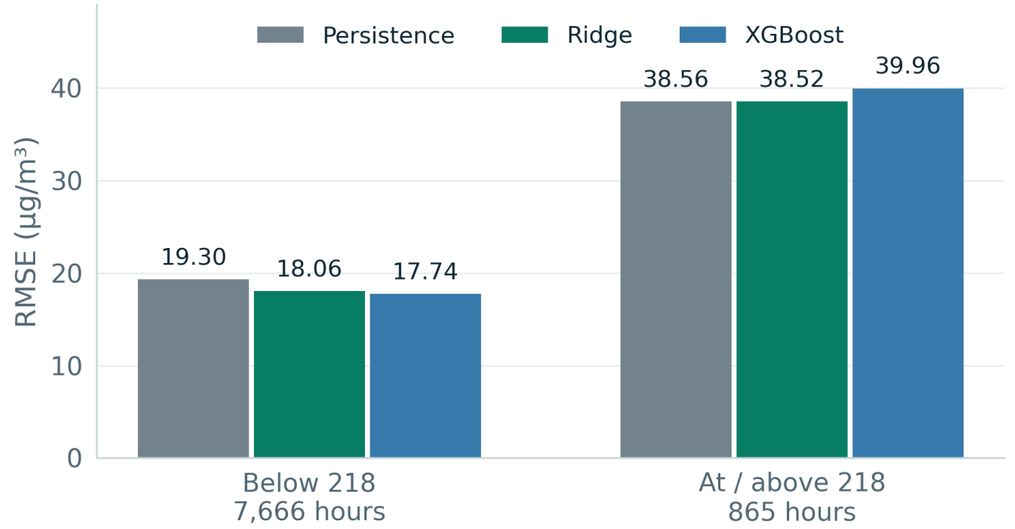
A single annual score hides important differences

Quarterly model rankings change, and XGBoost's pooled advantage does not extend to the high-pollution subset.

Quarterly holdout RMSE



Errors by observed pollution level



Both learned models beat persistence each quarter.

Ridge leads Q1 / Q3; XGBoost leads Q2 / Q4.
In Q1, XGBoost's MAE still exceeds persistence.

High-pollution RMSE: XGBoost 39.959 vs 38.557.

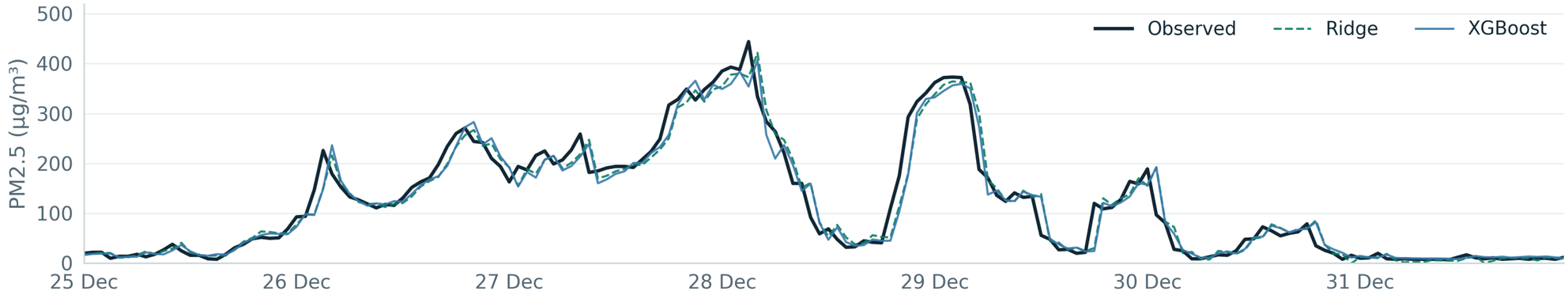
High-pollution bias: XGBoost −11.912; Ridge −13.313.
Both learned models underpredict this subset on average.

SOURCE: fold_metrics.csv and errors_by_pollution_regime.csv; high threshold fixed at training p90 = 218 µg/m³.

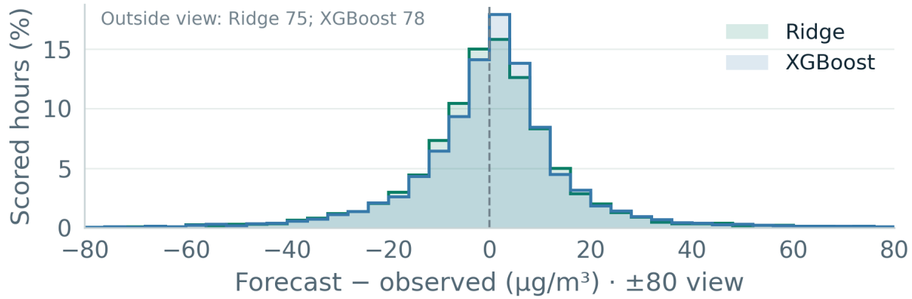
Look at the predictions, not just the summary

Saved rolling backtest forecasts · the final deployment refit is never used to generate these historical scores.

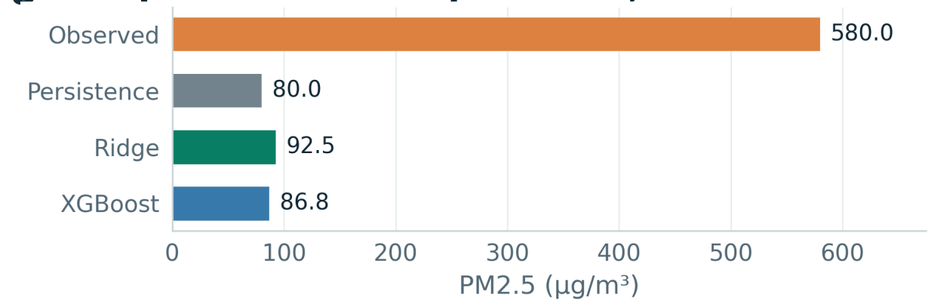
Actual versus predicted · final seven days of 2014



Signed residuals across the full holdout



Largest spike miss · 9 April 2014, 20:00



SOURCE: 2014 predictions.csv; trace shows 25–31 December; the spike is the largest XGBoost holdout miss.

From experiment artifacts to the Aircast demo

The interface invokes the original Python feature builder and saved pipelines; training and inference are separate.

INPUT
Manual observation / hourly CSV

→

VALIDATE
Flask + revalidated history

→

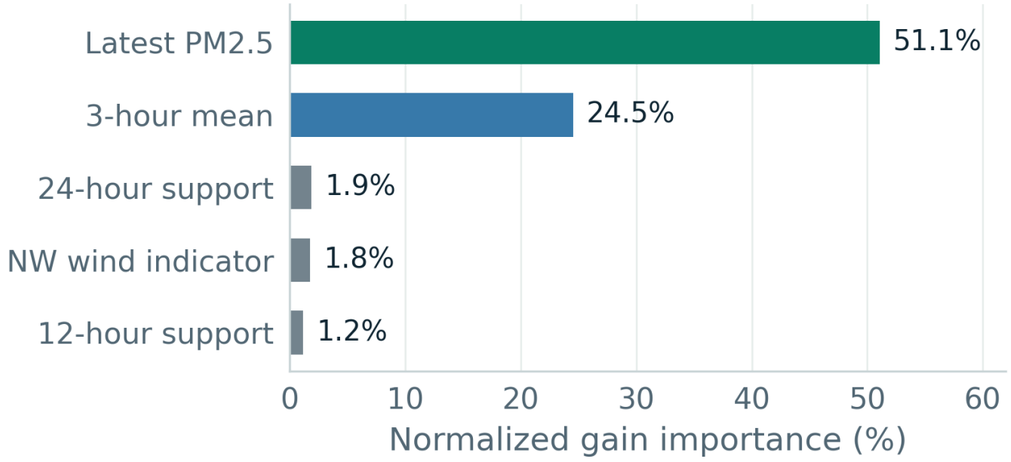
PREDICT
Saved Ridge / XGBoost pipeline

→

EXPLORE
History chart + forecast export

What the final XGBoost refit uses

Top five features · gain importance is not causation



Reproducibility and practical limits

17,062 **refitted predictions reproduced**

Maximum numeric difference < 6e−14 in the recorded audit.

Retained: config, hashes, versions, fold audits, predictions, cleaning exports, feature dictionary, CLI / notebook and tests.

Limits: historical Beijing; peak errors; assumed data availability.
Next: fresh evaluation, peak-focused methods and uncertainty.

Demo scenarios are historical. No calibrated intervals or AQI mapping.

SOURCE: Saved deployment-refit gain importance and the recorded historical/local prediction audit.

Thank you.

Questions & discussion

TEAM MEMBER	ENROLLMENT NUMBER
Samarth Khera	2401010407
Pratham Malhotra	2401010343
Adityaraj Pal	2401010042
Tattva Rajput	2401010478
Himanshu	2401010188

Phase 1: a reproducible pipeline, modest pooled gains, and clear evidence for the next experiment.